

OLLAMA VOICE ASSISTANT DOCUMENTATION

The process to achieve a functional voice assistant was successful but highlighted some performance challenges. There were 3 core technical problems: first, fixing the Speech-to-Text (STT) component by installing flac and adjusting time limits to prevent the microphone from transcribing the AI's own output; second, resolving long Ollama (LLM) processing latency by setting a 120-second timeout; and third, overcoming repeated Git conflicts during the submission phase.

Another issue was the poor quality and challenging default male voice of the espeak Text-to-Speech engine. The default voice was often distorted and difficult to understand which required updating the TTS parameters in the Python script to use a female voice (en+f3) and adjust the rate and pitch to improve clarity. The female voice was better but still difficult to understand ([see recording here](#))

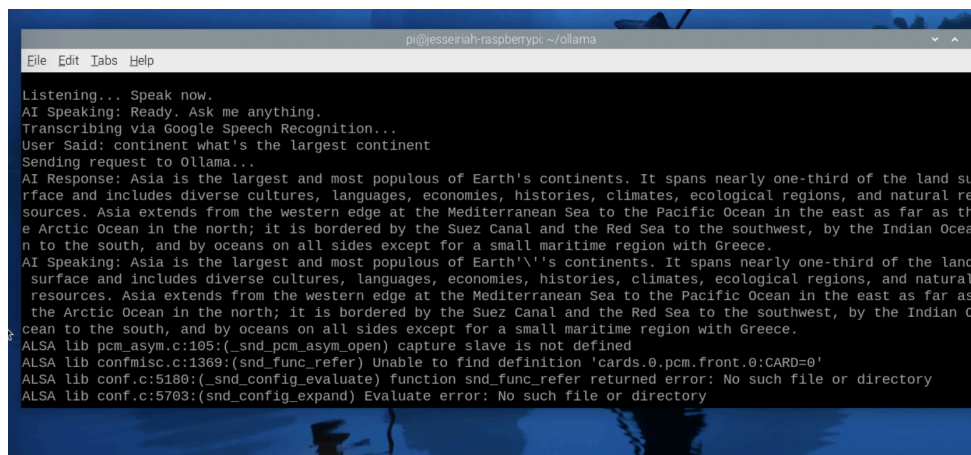
Baseline AI Response:

The very first successful response revealed the model's default behavior. After the system accidentally transcribed its own "Ask me anything" prompt, the Phi-3:mini model generated a lengthy, boilerplate text detailing its privacy policy and ethical limitations. However, the response was too long and formal for a practical voice assistant.

```
Listening... Speak now.  
AI Speaking: Ready. Ask me anything.  
Transcribing via Google Speech Recognition...  
User Said: ask me anything  
Sending request to Ollama...  
^CTraceback (most recent call last):
```

User Response to Performance:

Functionally, the system later achieved a successful full-loop dialogue when the user asked, "What's the largest continent?". This confirmed that the local Phi-3:mini model could process and respond to a complex query. However, the critical takeaway from the testing phase was the user's response to latency. The long wait time (30+ seconds for the first query) while the model processed the request caused a little anxiety and a loss of confidence in the system. Future designs should prioritize providing immediate verbal feedback ("I'm thinking now...") to manage user expectations and mask the unavoidable processing delay for conversational outputs.



```
pi@yesserah-raspberry: ~/ollama  
File Edit Tabs Help  
Listening... Speak now.  
AI Speaking: Ready. Ask me anything.  
Transcribing via Google Speech Recognition...  
User Said: continent what's the largest continent  
Sending request to Ollama...  
AI Response: Asia is the largest and most populous of Earth's continents. It spans nearly one-third of the land surface and includes diverse cultures, languages, economies, histories, climates, ecological regions, and natural resources. Asia extends from the western edge at the Mediterranean Sea to the Pacific Ocean in the east as far as the Arctic Ocean in the north; it is bordered by the Suez Canal and the Red Sea to the southwest, by the Indian Ocean to the south, and by oceans on all sides except for a small maritime region with Greece.  
AI Speaking: Asia is the largest and most populous of Earth's continents. It spans nearly one-third of the land surface and includes diverse cultures, languages, economies, histories, climates, ecological regions, and natural resources. Asia extends from the western edge at the Mediterranean Sea to the Pacific Ocean in the east as far as the Arctic Ocean in the north; it is bordered by the Suez Canal and the Red Sea to the southwest, by the Indian Ocean to the south, and by oceans on all sides except for a small maritime region with Greece.  
ALSA lib pcm_asym.c:105:(snd_pcm_asym_open) capture slave is not defined  
ALSA lib confmisc.c:1369:(snd_func_refer) Unable to find definition 'cards.0.pcm.front.0:CARD=0'  
ALSA lib conf.c:5180:(snd_config_evaluate) function snd_func_refer returned error: No such file or directory  
ALSA lib conf.c:5703:(snd_config_expand) Evaluate error: No such file or directory
```